



University of Moratuwa
Faculty of Computer Science and Engineering

Guest Review Intelligence & Action Management System

A Reputation-Protection Platform for The Kingsbury, Colombo

Final Module Report

*Turning thousands of scattered guest reviews into a prioritised,
department-owned action list, before problems define the brand.*

Prepared for



The Kingsbury, Colombo

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A working proof-of-concept built on live analysis of real Kingsbury reviews crawled from Google, TripAdvisor and Booking.com, augmented with engineered incident fixtures. Hotel financial figures are illustrative and explicitly flagged.

Abstract

A five-star hotel's reputation is no longer written only by its front desk. Increasingly it is mediated by algorithms, such as search rankings and conversational AI assistants, that read guest reviews and distil them into a one-line verdict. When a recurring complaint is repeated by enough guests it becomes a permanent caveat in how the property is described on every query. The Kingsbury, Colombo, a five-star hotel rated 4.0/5 and ranked #32 of 103 hotels in Colombo on TripAdvisor, illustrates the problem. Thousands of reviews are spread across multiple platforms, far beyond what any operations team can read, cross-reference and act on consistently. This report presents a working proof-of-concept that runs the same mechanism that judges the hotel, but *for* the hotel. A multi-platform ingestion layer feeds a natural-language analysis pipeline that scores every review for sentiment and a 0–100 reputation-risk value. A three-pass large-language-model pipeline (Google Gemini 2.5 Flash) then distils negative and mixed reviews into a small set of concrete, evidence-grounded, department-owned operational issues while preserving the guest's exact specifics such as room number, amount and time. Across an analysed corpus of 2,939 reviews (weighted toward the reputation-relevant negative and mixed reviews), 2,294 problem reviews collapsed into 117 active issues, roughly a 20:1 reduction, alongside 587 single-mention early-warning signals, at a marginal detection cost of about US\$0.25 per full run. The report situates the system commercially through a market analysis, a five-part feasibility study, a strategic risk analysis, and a pricing model sized for a Sri Lankan property. We conclude that even on conservative assumptions the platform protects a multiple of its cost, and we outline the path to production through incremental detection and property-management-system integration.

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1 Introduction

1.1 Background

Travellers no longer scroll through thousands of reviews before choosing a hotel. They ask a search engine or an AI assistant a direct question, such as *“Is The Kingsbury Colombo a good hotel?”*, and receive a synthesised answer drawn from exactly those reviews. That answer typically carries a caveat along the lines of *“well-regarded, though some guests mention dated rooms, noise from live music, and occasional service inconsistency.”* The caveat is not an opinion. It is a problem that surfaced in reviews, was repeated by enough guests, and became baked into how the property is described on every subsequent query. Online reputation has therefore become a direct commercial asset. Cornell’s analysis of lodging performance found that a one-percent improvement in a hotel’s online reputation score is associated with a 1.42% increase in revenue per available room (RevPAR) (Anderson, 2012), and more than 80% of travellers read reviews before booking (Tripadvisor, 2019; ReputationDefender, 2023).

1.2 Motivation

The mechanism that now judges the hotel, an AI reading its reviews, is precisely the mechanism a hotel should run *for itself*, but sharper and earlier. A generic assistant only reports that an issue has *already* bubbled up, and by then it is a public caveat that no tool can delete. The opportunity is to detect each operational fault at its *first* mention rather than its two-hundredth, route it to the responsible department, and fix the cause before it spreads. This is not review deletion or damage repair. It is a standing investment in the reputation the hotel has not yet lost.

1.3 Problem statement

A high-volume property generates more reviews, across more platforms and languages, than any operations team can read, de-duplicate, risk-rank and assign consistently. The cost is not the reading time. It is the *pattern that gets missed*, the single complaint today that is silently the two-hundredth of a recurring fault. The core problem this project addresses is the **cognitive-load and synthesis gap**, namely converting a continuous, multi-platform stream of unstructured reviews into a short, prioritised, evidence-backed and department-owned list of actionable issues.

1.4 Objectives

- O1.** To **ingest and normalise** guest reviews from multiple platforms (Google, TripAdvisor, Booking.com) into a single canonical store through modular, official-shaped connectors.
- O2.** To **analyse** each review for sentiment, owning department, and a transparent 0–100 reputation-risk score.
- O3.** To **detect operational issues** by distilling negative and mixed reviews into a small set of concrete, de-duplicated, evidence-grounded issues that preserve the guest’s specifics.
- O4.** To **prioritise and assign** each issue to a department with risk-based ranking, recurrence tracking, and human-in-the-loop resolution.
- O5.** To **evaluate the commercial case** through market, feasibility and risk analysis against a real five-star property.

1.5 Scope

The deliverable is a functioning proof-of-concept (POC) running locally, comprising a web application, an API, and a relational database. It serves four operator roles, namely an

administrator, an operations manager, a department head, and an analyst, each with a tailored view of the same data. It uses official-shaped connectors with sample and crawled data rather than live production credentials, and is sized for a single property. Production concerns such as live platform authorisation, incremental detection at full scale, and property-management integration are designed for but deliberately out of POC scope, and are discussed in Sections 8 and 13.

2 Market Analysis

2.1 Current approaches and competitors

Hotels manage reviews today in three broad ways, and each leaves a gap. **Manual triage**, where staff read reviews platform by platform, does not scale, drifts in consistency, and never reliably connects one complaint to the two hundred like it across a year. **Generic reputation and social-listening tools**, including established suites such as Medallia/ReviewPro, TrustYou and Revinate, aggregate reviews and present sentiment dashboards and word clouds, but they report broad themes such as “people mention food” rather than specific, owner-assignable incidents such as “breakfast buffet ran empty by 9 am”, and they attach no department ownership or tracking. **Native platform AI and general assistants** now summarise reviews directly, which is convenient for a traveller but reactive for a hotel, because by the time a theme is prominent enough to be summarised it is already a public caveat. Academic work on aspect-based sentiment analysis (Pontiki et al., 2016) similarly yields aspect categories rather than the concrete work orders an operations team can close.

2.2 Demand and review context

A reasonable question is whether a property of this size even generates enough review pressure to warrant a dedicated tool. Table 1 sizes the hotel from its real room count of 229 and clearly-flagged illustrative operating assumptions. At a representative 72% occupancy and a US\$130 average daily rate, the property turns roughly 60,000 occupied room-nights and about US\$7.8 M of room revenue per year, and at an estimated average stay of about 2.5 nights it sees on the order of 65 arrivals per day. Against its roughly 20,000 cumulative public reviews, current review velocity is on the order of 10 to 20 new reviews per day across platforms. In Asia, online travel agencies (OTAs), the commission booking channels such as Booking.com, account for roughly 70% of *online* hotel bookings (Digital Travel APAC / Cambodge Mag, 2024), and more than 80% of travellers read reviews before booking on *any* channel (Tripadvisor, 2019), so review pressure shapes a large share of demand for a property like this. These are not large numbers to *read*, which is the heart of a natural objection that a staff member could simply read them. But reading is not the task. The task is *synthesis*, namely connecting one complaint today to the two hundred spread across twelve months, three platforms and several languages, scoring each consistently, and spotting the one emerging issue among hundreds.

Table 1: Demand context for The Kingsbury. The room count of 229 is real, and operating ratios are illustrative benchmarks, flagged as such.

Quantity	Estimate
Rooms (actual)	229
Assumed occupancy (illustrative)	72%
Assumed average daily rate (illustrative)	US\$130
Occupied room-nights / year	≈ 60,200
Room revenue / year (illustrative)	≈ US\$7.8 M
Arrivals / day (at ~2.5-night stay)	≈ 65
Cumulative public reviews (all platforms)	≈ 20,000
Estimated new reviews / day	≈ 10–20

2.3 Market gap and positioning

The gap is the *action layer*. No mainstream approach combines hotel-specific issue detection, evidence and specifics down to the room number, department ownership and tracking, and early-warning detection of single-mention signals, at a cost a single property can absorb. That combination is the position this system takes.

3 Proposed Solution

3.1 Overview

The system is a three-tier application (Figure 1). A **Next.js / React** web client presents the operator screens. A **FastAPI** service exposes a REST API and houses the ingestion connectors, the natural-language analysis, and the issue-detection pipeline. State is persisted in **PostgreSQL**. Heavy natural-language models run locally, covering sentiment, zero-shot department classification, and sentence embeddings. Only the issue-detection reasoning calls an external large language model (Google Gemini 2.5 Flash), with a deterministic offline *stub* provider available for development and testing.

3.2 Key features

The product surfaces four capabilities. **One inbox for everywhere** unifies Google, TripAdvisor and Booking.com reviews in a single view. **AI issue detection** distils complaints into concrete, named operational issues. **Reputation-risk scoring** rates every review and issue from 0 to 100 for urgency. **Owned actions** route each issue to a department, prioritise it, and track it to resolution.

3.3 Unique value proposition

- Turns thousands of scattered complaints into one prioritised, owned and evidenced action list.
- Preserves the guest’s exact specifics, so an issue reads as “room 1413, US\$60 unexplained charge”, not “billing problems”.
- Catches an emerging problem at its first mention rather than its two-hundredth.
- Runs at near-zero marginal cost, about US\$0.25 for a full detection pass.
- Keeps a human in the loop, so the system proposes and staff decide.

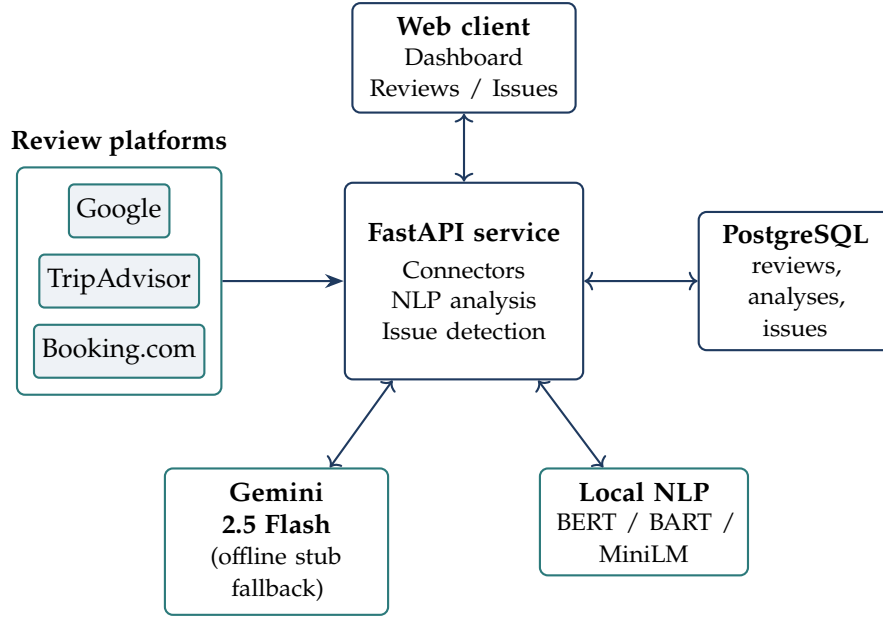


Figure 1: System architecture. Reviews flow from modular platform connectors into the FastAPI service, which analyses them with local NLP models and an external LLM, persists results in PostgreSQL, and serves the operator-facing web client.

4 System Design and Methodology

4.1 Ingestion and the canonical data model

Each platform has a dedicated **connector** that accepts the platform’s own (“official-shaped”) payload and maps it to a canonical review. No scraping is required, and a source can be swapped without touching the analysis engine. Raw payloads are preserved verbatim, with a content hash for idempotent and duplicate-safe ingestion, and then normalised. The data model is a linear enrichment pipeline. A raw review becomes a normalised review, which gains a one-to-one analysis record, and may be linked as evidence to one or more detected issues. The full schema is shown in Figure 8.

4.2 Natural-language analysis

Every normalised review is scored by three local models. The first is a multilingual BERT sentiment classifier (positive, mixed or negative). The second is a zero-shot department classifier built on BART-MNLI (Lewis et al., 2020) over six hotel departments (Front Office, Housekeeping, Food & Beverage, Engineering, Guest Relations, Management). The third is a MiniLM sentence-embedding model (Reimers and Gurevych, 2019) that produces a 384-dimensional vector used later for de-duplication and issue clustering.

Reputation-risk score. Each review receives a transparent, auditable 0–100 reputation-risk score, labelled low, medium, high or critical, computed as a bounded sum of interpretable factors, so a manager can see *why* a review is urgent.

$$\text{risk} = \min\left(100, \underbrace{\frac{5-r}{4} \cdot 30}_{\text{rating}} + \underbrace{(-s) \cdot 25}_{\text{sentiment}} + \underbrace{w_d}_{\text{dept.}} + \underbrace{u}_{\text{urgency}} + \underbrace{v}_{\text{visibility}} + \underbrace{\rho}_{\text{recency}} + \underbrace{\delta}_{\text{duplicate}}\right) \quad (1)$$

Here r is the star rating, $s \in [-1, 1]$ the sentiment score, and w_d a per-department weight (Management highest). The term u adds points for urgency words such as “refund”, “unsafe”

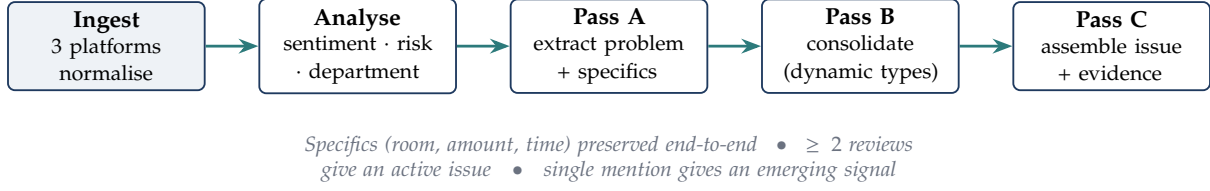


Figure 2: The end-to-end pipeline. Reviews are ingested and analysed, then a three-pass LLM stage extracts, consolidates and assembles concrete, evidence-grounded issues.

or “manager”, v for public visibility through helpful votes, a public URL or unanswered status, ρ for recency, and δ for a detected duplicate. Every contributing factor is stored alongside the score for audit.

4.3 Issue detection through a three-pass LLM pipeline

The core contribution is the issue-detection pipeline (Figure 2), which runs only over **negative and mixed** reviews, the reputation-relevant ones, in three passes.

Pass A, Extract.

Reviews are sent to the LLM in small batches, and for each it returns a short problem summary, the owning department, and the concrete *specifics* such as room or floor, item, amount and time.

Pass B, Consolidate.

All extracted problem labels are clustered into a *dynamic* taxonomy of canonical issue types rather than a hard-coded list. Genuinely distinct severe incidents such as a pest sighting, theft or a safety hazard are deliberately kept separate rather than folded into a generic category.

Pass C, Assemble.

Each canonical type becomes one issue with an evidence-grounded description. A type supported by ≥ 2 reviews becomes an **active** issue, while a single-mention type becomes an **emerging** early-warning signal. A stable `cluster_key` lets the pipeline be re-run idempotently while preserving manual state such as assignee and resolved status.

The defining property is **specificity preservation**. The output is not “billing problems” but “room 1413, US\$60 unexplained charge”, a closeable work order rather than a theme.

4.4 Data and evaluation methodology

The analysed corpus comprises **2,939 reviews**. These combine *real reviews crawled from Google and TripAdvisor* for The Kingsbury with a set of *engineered incident fixtures*. The corpus was deliberately weighted toward **negative and mixed** reviews because those are the only reputation-actionable inputs. The detection pipeline reads negative and mixed reviews exclusively, and positive reviews yield no issues to act on. The engineered fixtures were generated from a library of specific incident scenarios, each carrying concrete facts such as room, floor, item, amount and time, in order to **stress-test specificity preservation**, because real crawled reviews frequently omit such details. On authenticity, two of the three sources are *verified-stay channels*, since on Booking.com and predominantly on TripAdvisor reviews come from guests with a booking, which constrains fabrication, whereas Google is the open channel and is treated as lower-confidence. The system also never acts on a single unverified post, because an issue is meaningful through the *aggregation* of many independent reviews, and a human always resolves or dismisses it. The legal and data-protection dimension of this is examined in Section 7.4.

5 Implementation

5.1 Main interfaces

The web client presents three working screens for the operations team. The Dashboard (Figure 3) is the at-a-glance product surface. The Reviews page (Figure 4) is the filterable stream of analysed reviews. The Issues page (Figure 5) is the ranked action queue, and opening any issue reveals the detail side-sheet (Figure 6) with its linked reviews, evidence snippets and event history. A separate Emerging candidates tab (Figure 7) lists the single-mention early-warning signals.



Figure 3: The Dashboard, a product showcase of the operations overview.

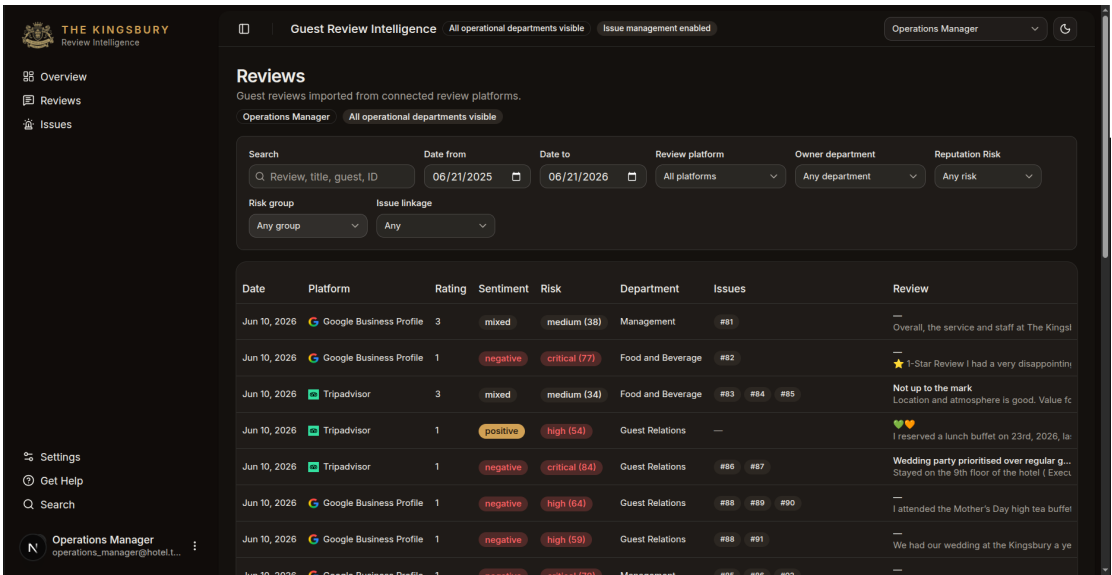


Figure 4: The Reviews page, with per-review sentiment, reputation risk, department and issue links.

Issue	Department	Status	Priority	Risk	Recurrence	Last seen	Assignee
Staff clueless about service Twenty-two guest reports indicate significant staff training iss...	Food and Beverage	active	urgent	89	22	Jun 10, 2026	—
Pool closed without prior notice or is dirty On New Year's Eve, a guest at Honey Beach Club reported the	Engineering	active	urgent	89	43	Jun 8, 2026	—
Manager unresponsive or unhelpful to guest compl... Guests reported 72 instances of management being unrespon...	Management	active	urgent	87	72	Jun 10, 2026	—
Long wait for food or drink service Guests experienced significant wait times for food and drink s...	Food and Beverage	active	urgent	87	40	Jun 10, 2026	—
Reading lamp or electric kettle is broken A guest in room 824 on the third floor reported old and damag...	Engineering	active	urgent	87	59	Jun 10, 2026	—
Room lacks basic amenities or sufficient drinking ... Multiple guests reported insufficient drinking water, with 23 tot...	Housekeeping	active	urgent	87	23	Jun 10, 2026	—
Carpets or furniture in room are stained or dirty Nine guest reports indicate significant issues with stained and	Housekeeping	active	urgent	87	9	Jun 10, 2026	—
Food delivered cold or late from room service Room service is experiencing significant issues with timely anc...	Food and Beverage	active	urgent	87	22	Jun 10, 2026	—
Room is dusty or dirty							

Figure 5: The Issues page, the active-issue queue ranked by reputation risk, with status, priority and assignee. An “Emerging candidates” tab holds the 587 single-mention signals.

Issue	Department	Status
Staff clueless about service Twenty-two guest reports indicate significant staff training iss...	Food and Beverage	active
Pool closed without prior notice or is dirty On New Year's Eve, a guest at Honey Beach Club reported the	Engineering	active
Manager unresponsive or unhelpful to guest compl... Guests reported 72 instances of management being unrespon...	Management	active
Long wait for food or drink service Guests experienced significant wait times for food and drink s...	Food and Beverage	active
Reading lamp or electric kettle is broken A guest in room 824 on the third floor reported old and damag...	Engineering	active
Room lacks basic amenities or sufficient drinking ... Multiple guests reported insufficient drinking water, with 23 tot...	Housekeeping	active
Carpets or furniture in room are stained or dirty Nine guest reports indicate significant issues with stained and	Housekeeping	active
Food delivered cold or late from room service Room service is experiencing significant issues with timely anc...	Food and Beverage	active
Room is dusty or dirty		

Pool closed without prior notice or is dirty Close

Engineering

On New Year's Eve, a guest at Honey Beach Club reported the pool area was dirty at 7 AM. Another guest on day 6, in room 981 on the 5th floor, found the pool cloudy with an unusually high amount of leaves. There was also a report of too much chlorine substance in the pool water.

active Risk: 89 Priority: urgent Recurrence: 43x

pool day area 981 5th floor honey

First seen: Jun 17, 2025 Last seen: Jun 8, 2026

Resolved: — Recurred: —

Assignee: — Title by: gemini

Mark Resolved

Linked Reviews (43)

Tripadvisor · jayft3 · ★ 1 · Jun 8, 2026 Trigger

We are German Citizens and started our holiday in Sri Lanka. I and my family checked in the Kingsbury Hotel on the 7 August, 3 Exec.rooms 902, 904, 906, for 3 nights.

1. we have experienced only terrible things at the hotel. loud music from the pool area goes till 1 am. Dinner restaurants staff are untrained. Not greeting guests.

2. Can not use the pool facilities at 7am in the morning because pool is dirty from previous night parties. no towels available, no greetings from any managers.

pool tiles are broken and it is danger ...

3. High Tea at the Lobby area ... staff are terrible. only looking for rich and famous to serve. We had to pay lot of money even we had our valid gift vouchers. No body is taking a responsibility. every one makes mistakes. And even have to pay for their mistakes.

Figure 6: The issue detail side-sheet, showing the linked reviews, evidence snippets and the event history for one issue.

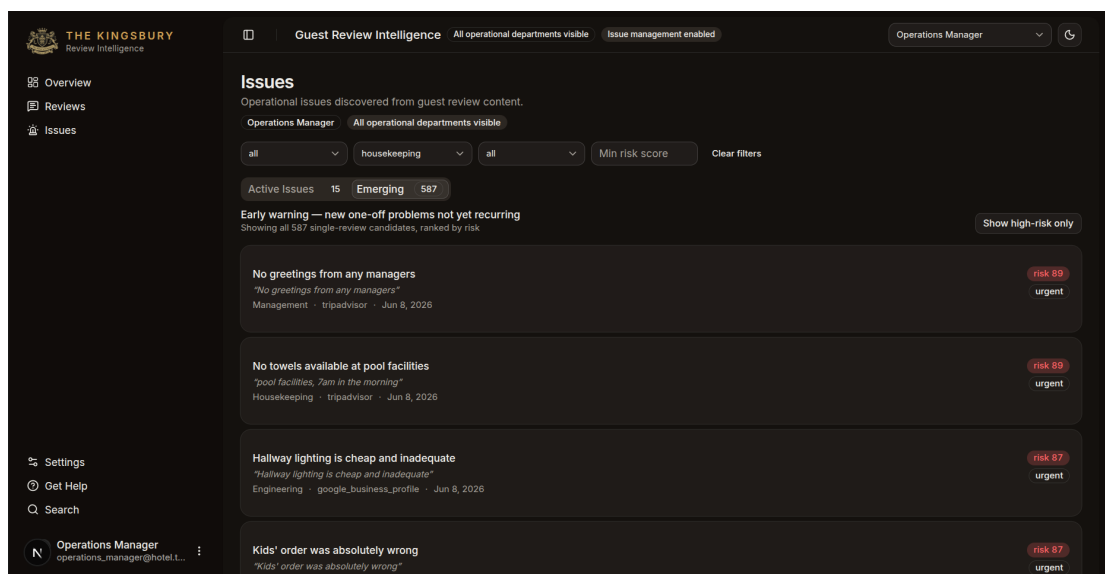


Figure 7: The Emerging candidates tab, listing single-mention early-warning signals ranked by reputation risk.

5.2 Database schema

The relational schema (Figure 8) preserves provenance from the raw provider payload through to a department-owned issue, with every issue traceable to the exact reviews and snippets that evidence it.

5.3 Key interaction flows

Figure 9 traces a review from a platform connector through analysis to a detected issue, and Figure 10 traces an operations manager triaging and resolving an issue.

5.4 Important implementation decisions

Three decisions shaped the build. First, **the heavy NLP runs locally** on commodity hardware, which keeps per-review analysis free and private, while only the issue-detection reasoning uses an external LLM. Second, **Gemini 2.5 Flash with an offline stub fallback** gives high-quality detection at about US\$0.25 per run, with a deterministic provider for tests and for running without a key. Third, **a stable cluster_key** makes the full rebuild idempotent, so the pipeline can be re-run while preserving manual state such as assignee and resolved status.

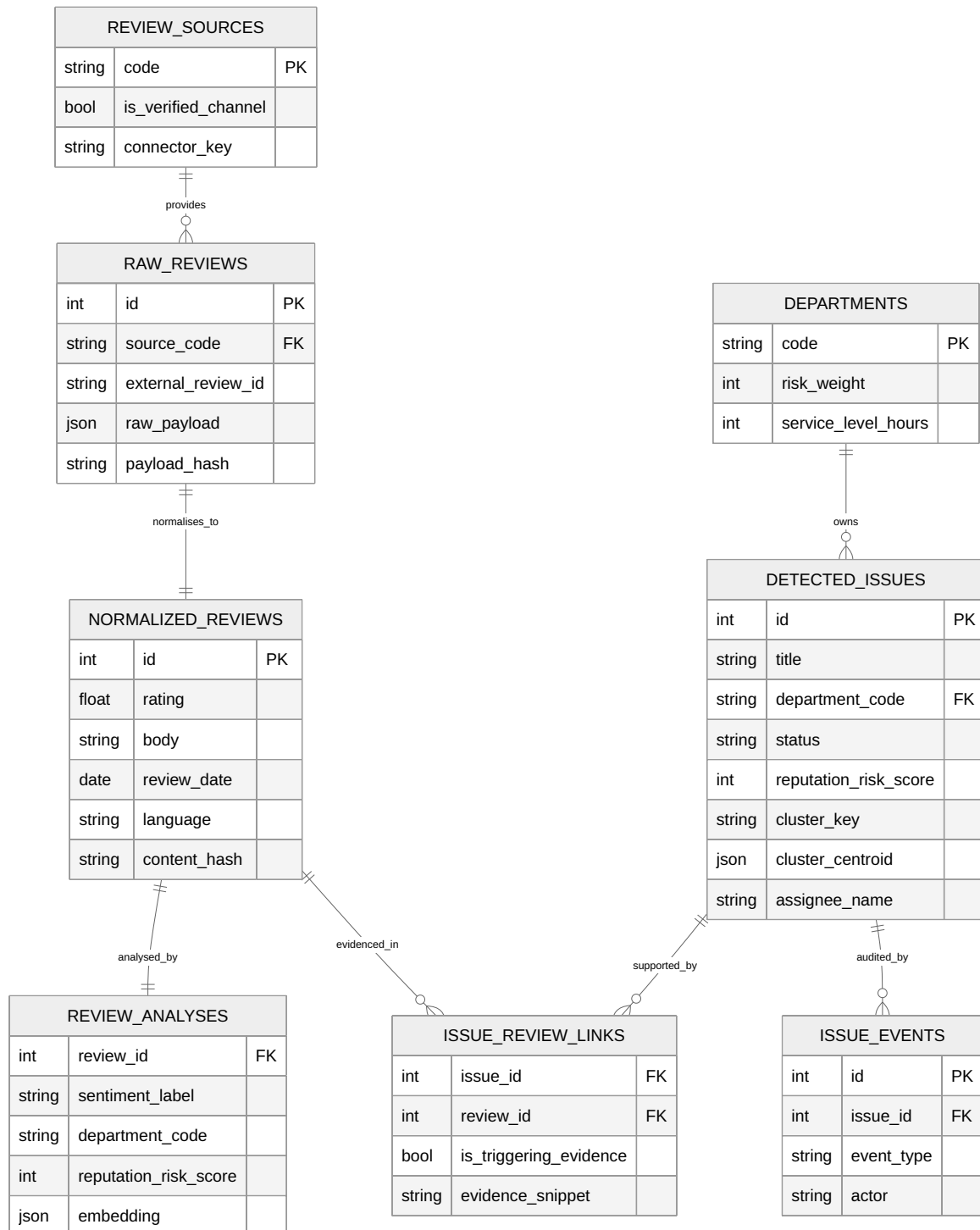


Figure 8: Database schema. The enrichment chain runs from raw_reviews to detected_issues, joined to the reviews that evidence them through issue_review_links.

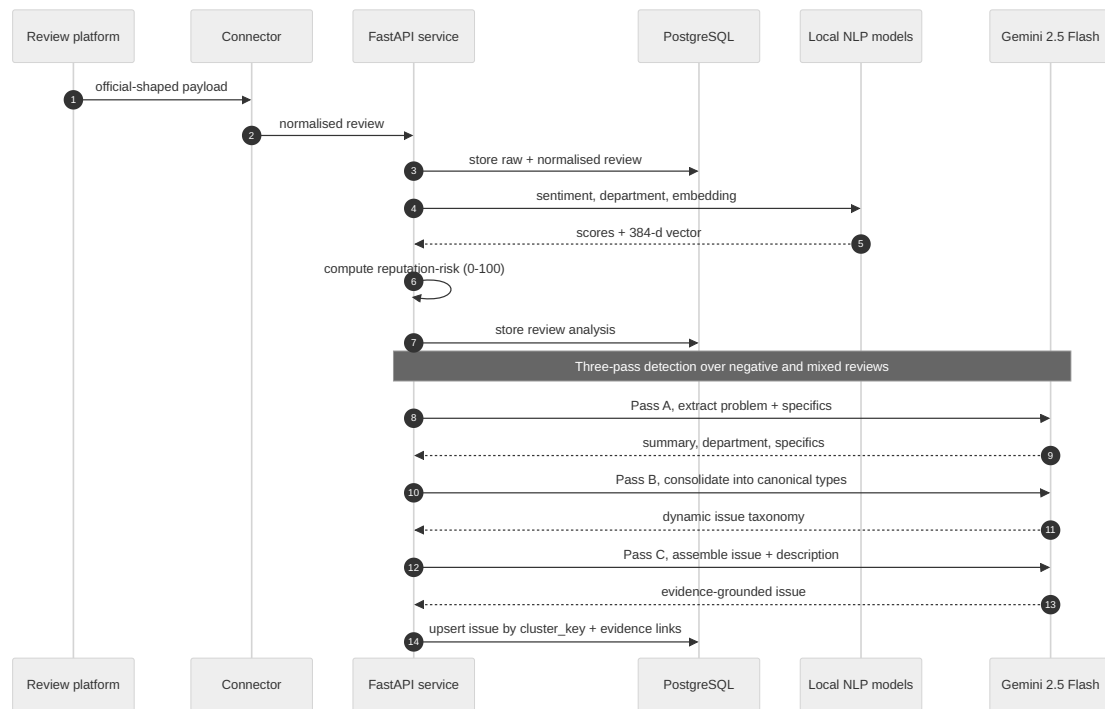


Figure 9: Detection flow. A review is ingested, scored by the local models, and passed through the three LLM passes before issues are upserted by `cluster_key`.

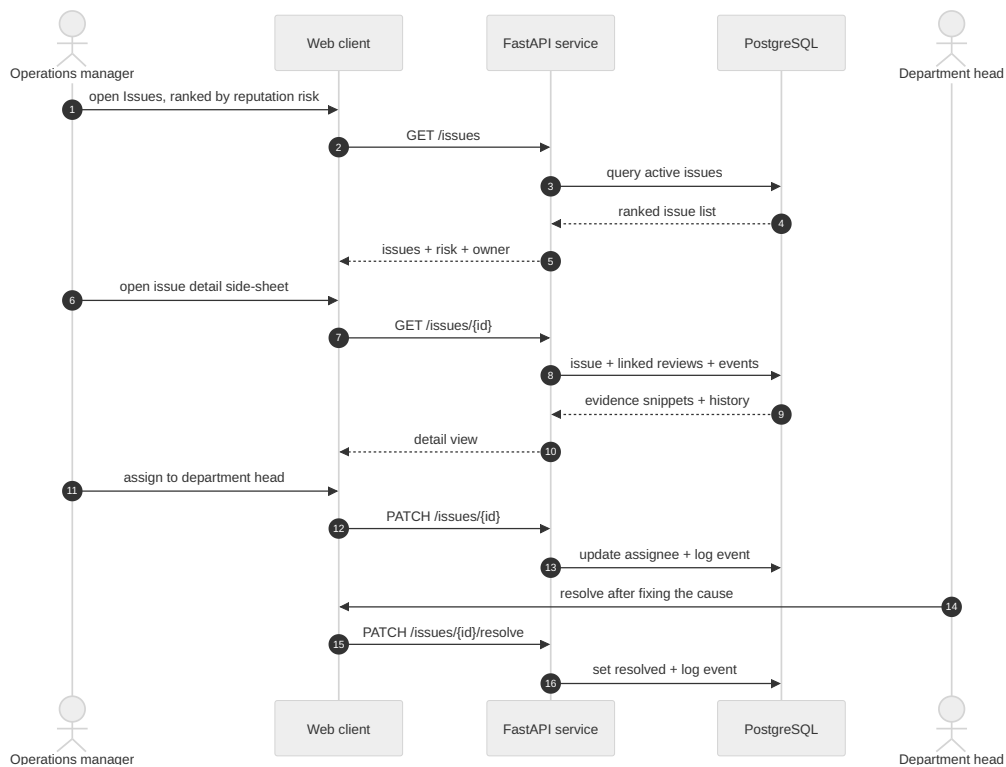


Figure 10: Resolution flow. An operations manager opens the ranked issues, inspects the evidence, assigns to a department head, who resolves once the cause is fixed.

6 Results and Findings

6.1 Scale and compression

Table 2 summarises the headline result. The 2,294 negative and mixed reviews collapsed into 117 active issues, a roughly **20:1 compression**, so that managers act on 117 things rather than reading thousands of complaints. A further 587 single-mention *emerging* signals were caught as early warnings, and 77% of complaint reviews were traced to a concrete, fixable issue via 3,793 evidence links.

Table 2: Corpus and detection outcomes.

Metric	Value
Reviews analysed (negative/mixed-weighted corpus)	2,939
Negative & mixed (reputation-relevant) reviews	2,294
Active issues detected	117
Compression ratio (problem reviews to active issues)	≈ 20:1
Emerging (single-mention) early-warning signals	587
Complaint reviews traced to a concrete issue	77% (3,793 links)
Issues at critical risk (≥ 80)	54
Marginal LLM cost per full detection run	≈ US\$0.25

6.2 The issues that matter

Table 3 lists the highest-impact detected issues. The single most-complained-about problem, loud music and event noise with 201 mentions at risk 89, is *exactly* the caveat a conversational AI repeats about the hotel, which confirms that the system measures the same signal the market already reads. Issues arrive with receipt-level specificity, so billing complaints are not a generic category but, for example, “room 1413, US\$60 unexplained charge”, a work order the front office can close immediately. Active issues distribute across owners as Food & Beverage 37, Front Office 25, Engineering 25, Housekeeping 15, Management 9 and Guest Relations 6, for 117 in total, giving each department head a self-contained, ranked queue.

Table 3: Top detected issues (verbatim from the system).

Issue	Mentions	Risk	Owner
Loud music / event noise disturbance	201	89	Management
Air conditioning fails to cool rooms	174	74	Engineering
Long wait at check-in / check-out	169	85	Front Office
Breakfast buffet not replenished	136	72	Food & Beverage
Incorrect / unexplained charges on bill	122	82	Front Office
Slow / unstable WiFi	114	85	Engineering
Poor food quality	113	85	Food & Beverage

7 Feasibility Study

7.1 Financial feasibility

The platform is inexpensive to run. The dominant marginal cost is the LLM detection run at about US\$0.25 over roughly 1,000 reviews, and even run daily that is under US\$100 per year.

Local NLP models run on commodity hardware at no per-use cost, and infrastructure for a single property is modest.

Pricing. The product is positioned as a one-time licence plus a small support fee, sized for a single Sri Lankan property rather than an enterprise buyer (Table 4).¹ The hotel pays a one-time product and implementation licence, a small monthly support fee, and the very low runtime cost directly. An alternative we considered is an annual SaaS fee of about LKR 240,000 (around US\$800) with runtime folded in, which is simpler to budget and better for a multi-property rollout, listed here as the secondary option.

Table 4: Proposed pricing under the recommended model, with indicative figures.

Component	LKR	US\$
One-time product licence + implementation	350,000	≈1,150
Monthly support & maintenance	7,500 / mo	≈25 / mo
LLM runtime (paid by hotel, daily runs)	<15,000 / yr	<50 / yr
Year-1 total (illustrative)	≈440,000	≈1,450
Ongoing annual (support + runtime)	≈90,000	≈300

Revenue impact and return. A naive estimate applies the Cornell elasticity, where +1% reputation gives +1.42% RevPAR (Anderson, 2012), to the entire US\$7.8 M room revenue, giving about US\$111k per 1% of reputation protected. That overstates the effect, because not all revenue responds to online reputation, so we estimate the impact against only the revenue base that online reputation plausibly influences and present a range (Table 5). The lower bound attributes impact only to OTA-booked room-nights, taken as about 25% of total room-nights for a property with substantial direct and corporate business. The upper bound recognises that more than 80% of travellers read reviews before booking on any channel (Tripadvisor, 2019; ReputationDefender, 2023), so reputation shapes demand well beyond the OTA slice. Each 1% of reputation protected is therefore worth roughly US\$28k to US\$89k per year, that is about LKR 8.4 M to LKR 26.7 M, against a year-one cost near LKR 440,000. Even on the lower bound, protecting half a percent of reputation repays the year-one cost many times over, and the elasticity cuts both ways, so this is also the revenue at risk from a 1% erosion.

¹Figures are indicative and converted at approximately LKR300 per US dollar. Hotel financials elsewhere in this report are illustrative.

Table 5: Annual value of protecting 1% of online reputation, applying the Cornell RevPAR elasticity only to the share of room revenue that online reviews actually influence. All hotel financials are illustrative.

Scenario	Influenced share	Influenced revenue base (US\$)	Value of +1% (US\$ / LKR)
Lower bound (OTA bookings only)	≈25%	≈2.0 M	28k / 8.4 M
Middle estimate (all online bookings)	≈55%	≈4.3 M	61k / 18.3 M
Upper bound (all review-influenced demand)	≈80%	≈6.3 M	89k / 26.7 M

OTA means an online travel agency such as Booking.com, the commission channel a guest books through online, as opposed to direct website, corporate or walk-in bookings. The influenced share is the portion of the hotel's roughly US\$7.8 M annual room revenue whose demand is shaped by online reviews. The value of +1% is that influenced revenue base times the Cornell figure of 1.42% RevPAR per 1% of reputation, for example 1.42% of US\$2.0 M is about US\$28k, which is roughly LKR 8.4 M.

7.2 Technical feasibility

The system is built and running on a conventional stack of Next.js, FastAPI and PostgreSQL, with the heavy models (BERT sentiment, BART zero-shot classification, MiniLM embeddings) running locally and only the detection reasoning calling Gemini. The deterministic stub provider means the entire pipeline runs offline without an API key, which de-risks both development and demonstration. Nothing in the design requires specialised hardware beyond a commodity machine.

7.3 Resource and time feasibility

The proof-of-concept was delivered by a five-member student team within a single academic cycle using free-tier and open-source components, so no licence costs were incurred. The same modular structure that made the POC tractable, namely independent connectors, local models and a single detection service, keeps the path to a production pilot short.

7.4 Social and legal feasibility

Socially, the platform improves guest experience by helping a hotel fix recurring faults faster, and it fits existing operations as a ranked, owned action list rather than another raw feed. Legally, the design is built to respect the constraints that a review-driven product faces. It uses official-shaped, modular connectors rather than scraping, which keeps ingestion within platform terms of service. Personal data is minimised and redacted in the interface, and the system stores operational issues rather than guest profiles, which aligns it with Sri Lanka's Personal Data Protection Act and with the GDPR principles that apply to international guests. Because the system proposes and a human decides, and because every issue is evidence-linked, the legal exposure of acting on a fabricated review is low. The clearest dependency is on third-party platform terms, which is treated as a first-class risk in Section 8.

8 Risk and Strategic Analysis

We analyse the system as a product venture using four standard frameworks.

8.1 SWOT

Table 6: SWOT analysis of the system as a product.

Strengths	Specificity-preserving detection, where room, amount and time give real work orders. Multi-platform unification. Evidence-linked and confidence-scored with a human in the loop. Early-warning detection of 587 single-review signals. Near-zero marginal cost with a department-ownership workflow.
Weaknesses	Third-party data dependency with no official live API yet. A full-corpus LLM rebuild whose cost grows with volume. Possible LLM mis-merge or hallucination needing human checks. A heuristic reputation-risk score not yet validated against revenue outcomes. Cold-start and variable non-English quality.
Opportunities	AI-era reputation defence as guests decide via assistants. Chain and multi-property rollout. Closing the loop with PMS and ticketing and auto-drafted responses. Predictive analytics and cross-hotel benchmarking. More sources and languages.
Threats	Platform terms or API changes restricting review access. Funded incumbents such as Medallia/ReviewPro, TrustYou and Revinate. LLM-provider pricing or policy changes. Native platform AI summaries commoditising review insight. Privacy regulation and review fraud polluting the signal.

8.2 Porter's Five Forces

Table 7: Porter's Five Forces.

Force	Rating	Why
New entrants	Moderate	LLM APIs lower the barrier, but domain depth, specificity and the ownership workflow form a moat.
Supplier power	High	Dependence on review platforms and one LLM provider, mitigated by modular connectors and a provider-agnostic LLM layer with an offline fallback.
Buyer power	Moderate	Hotels have alternatives and low trial cost, but the embedded workflow creates stickiness once adopted.
Substitutes	High	Manual triage, generic listening tools, native platform AI and general assistants. This is the force to beat, answered by action-management and early warning rather than summaries.
Rivalry	Moderate–High	Established vendors exist, but few combine detection, ownership and early warning at this cost point.

8.3 PESTLE

Table 8: PESTLE macro-environment.

Factor	Direction	Implication
Political	Neutral	Tourism policy and stability drive hotel demand, and data-localization sentiment is rising.
Economic	Tailwind	Cost-pressured hotels find cheap automation attractive, and reputation ties directly to ADR and RevPAR.
Social	Tailwind	Guests increasingly decide via AI assistants and reviews, raising demand for reputation defence.
Technological	Mixed	Cheaper and better LLMs help, but native platform AI summaries threaten.
Legal	Headwind	GDPR, Sri Lanka PDPA, platform terms, PII handling and anti-manipulation rules impose a compliance burden.
Environmental	Neutral	Low direct impact, and it can surface sustainability complaints with a modest compute footprint.

8.4 Risk register and mitigations

Table 9: Top risks ranked by exposure (likelihood by impact) with mitigations.

Risk	L × I	Mitigation
Platform access or terms change cuts off review data	High High	× Modular, official-shaped connectors with no scraping, so a source can be swapped without touching the engine.
LLM cost explodes at scale	High Med	× Incremental detection, matching new reviews to stored cluster centroids and calling the LLM only for novel issues, so cost tracks new volume.
False positives erode trust	Med High	× Evidence-linked and confidence-scored issues with human resolve or dismiss.
Incumbent or native-AI competition	Med High	× Differentiate on action-management, specificity and early warning, not summarisation.
Privacy or PII non-compliance	Low High	× PII redaction in the UI, explicit data policy, no scraping, regional-law awareness.

8.5 Strategic verdict

The position is a viable and defensible niche. The system is strong on workflow, specificity and cost, and exposed on supplier power and substitutes, both of which were engineered around through modular suppliers and an action layer that summarisers do not replicate. The investment priorities are clear, namely incremental detection for scale, live integrations to close the loop, a provider-agnostic LLM layer, and ongoing compliance. The reputation-risk score is the strategic anchor that keeps the highest-damage issues surfaced first even as volume grows.

9 Project Timeline

The project ran across one academic cycle, from idea framing through design, development, evaluation and reporting (Figure 11). Development was sequenced so that ingestion and analysis were in place before the issue-detection pipeline, which in turn preceded evaluation on the assembled corpus.

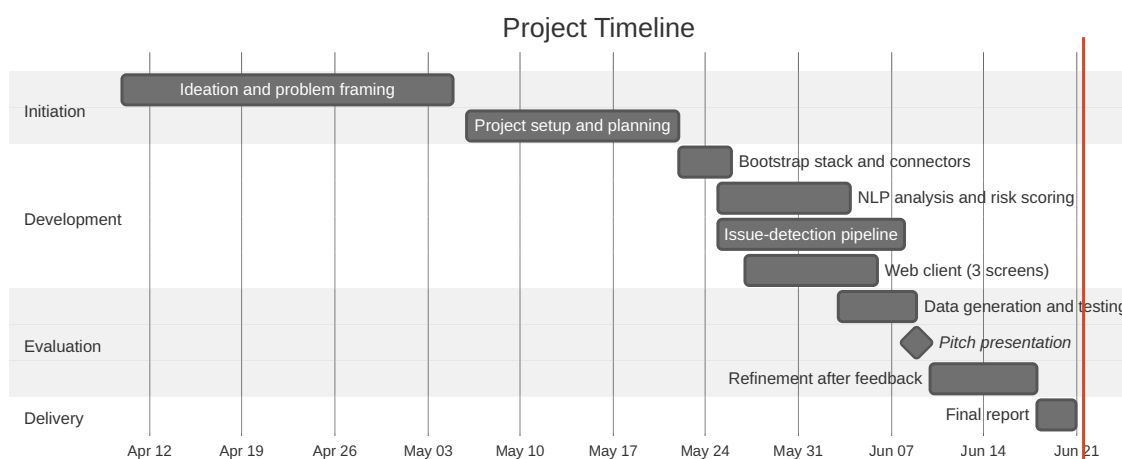


Figure 11: Project timeline across initiation, design, development, evaluation and delivery, from ideation in April 2026 to delivery on 21 June 2026.

10 User Journeys

The product is designed around a single recurring journey for the operations team. A guest leaves a review on any platform. The connector ingests it, the analysis scores it, and if it describes a recurring problem the detection pipeline attaches it to an issue, or raises a new emerging signal if it is the first of its kind. The operations manager opens the Issues page, already ranked by reputation risk, inspects the evidence in the detail side-sheet, and assigns the issue to the responsible department head. The department head fixes the underlying cause and marks the issue resolved, which is recorded in the issue's event history. The same flow is shown as an interaction sequence in Figure 10. A second journey serves the analyst, who uses the Dashboard and Reviews filters to monitor sentiment and risk trends and to confirm that resolved issues stop recurring.

11 Challenges Faced

The hardest problem was detection quality. An earlier embedding-and-clustering approach produced hundreds of vague, overlapping clusters that no manager could act on. Rebuilding it as the three-pass LLM pipeline, with a dynamic taxonomy and deliberate separation of severe distinct incidents, was what turned the output into a small set of clean, specific issues. Preserving specificity was itself a challenge, since a model summarising at scale tends to generalise away the room number and the amount, which is why those facts are extracted explicitly and carried through every pass. Controlling cost while keeping quality drove the choice of a fast, inexpensive model with an offline fallback. Finally, real crawled reviews often lacked the concrete details needed to test specificity, which is why the corpus was augmented with engineered incident fixtures.

12 Community and Industry Impact

Reputation is a shared asset for Sri Lanka's hospitality sector, where tourism is a major source of foreign earnings. A tool that helps a hotel detect and fix recurring faults early protects not only that property's revenue but also the guest experience that shapes the country's wider reputation as a destination. Because the cost structure is low, the same capability is within reach of smaller and independent hotels that cannot afford enterprise reputation suites, which helps level the field between large chains and local operators.

13 Lessons Learned and Future Work

Two lessons stood out. First, for review intelligence the value is in the action layer, not the summary, so designing around owned, evidenced, early action from the start was the decision that mattered most. Second, evaluating on data with concrete specifics early would have surfaced the specificity-preservation requirement sooner. The principal next steps are a full detection run over the hotel's complete real review history, the incremental-detection path that bounds LLM cost at production scale, property-management integration to corroborate complaints against actual folios and convert flagged issues into verified ones, and multi-property support to amortise the platform across a hotel group.

14 Conclusion

This report presented a working proof-of-concept that turns a high-volume, multi-platform stream of guest reviews into a short, prioritised, evidence-grounded and department-owned list of operational issues. On an analysed corpus of 2,939 reviews the system compressed 2,294 problem reviews into 117 active issues, a roughly 20:1 reduction, plus 587 early-warning signals, at a marginal cost of about US\$0.25 per run, while preserving the guest's exact specifics down to room numbers and billed amounts. The market, feasibility and risk analyses show a defensible niche with a low, realistic price for a Sri Lankan property and a revenue-protection value that dwarfs that price. A hotel cannot delete what an algorithm has already learned about its past, but for cents per analysis it can ensure tomorrow's problems never become the next permanent caveat.

A Annex: Individual Contribution Summary

The project was completed as an integrated team effort from initial problem discovery through to solution implementation and reporting. All team members contributed to brainstorming the hotel review intelligence concept, refining the scope around The Kingsbury case study, reviewing the final demo flow, and validating that the system remained an academic proof-of-concept rather than a claim of live production platform access. The table below records the main responsibility areas used to divide the work.

Table 10: Team member contributions from ideation to implementation.

Team member	Primary contributions
EKANAYAKA D.M.Y.N.B. (220153R)	Led the problem-framing and business-domain work that shaped the system as a reputation-protection tool rather than a generic review dashboard. Contributed to the initial brainstorming around how AI assistants and public reviews influence hotel reputation, helped define The Kingsbury case-study problem, and translated scattered guest complaints into the product concept of evidence-backed, department-owned operational issues. Prepared the commercial argument around reputation risk, review-driven revenue exposure, analyst time savings, pricing, ROI and future production value. Also contributed to the report sections on introduction, motivation, business analysis, limitations, conclusion and the final project narrative.
FERNANDO N.P.A. (220168R)	Led the backend architecture and API implementation. Designed the FastAPI service around modular, official-shaped review connectors for Google Business Profile, TripAdvisor and Booking.com, and implemented the ingestion path that preserves raw review payloads while creating normalized review records for analysis. Worked on the PostgreSQL schema, SQLAlchemy/Alembic migrations, seed data, idempotent import behaviour, review-analysis persistence, issue-review evidence links and REST endpoints used by the dashboard, review list and issue-management screens. Also prepared operational scripts for migration, import, fixture validation, database restore, demo execution and end-to-end verification.
PRASANNA W.A.S.D. (220488X)	Led the natural-language analysis and issue-detection implementation. Worked on the local NLP pipeline for sentiment classification, zero-shot department classification, sentence/review embeddings and the 0–100 Reputation Risk score used to prioritise operational damage. Designed and refined the specificity-preserving three-pass LLM detection flow: extracting concrete problems from negative and mixed reviews, consolidating repeated complaints into canonical issue types, and assembling evidence-grounded issue descriptions while retaining facts such as room, amount and time. Also contributed to active-versus-emerging issue logic, recurrence-ready cluster keys, issue evidence linking, confidence handling, and the provider-agnostic LLM client using Gemini with an offline stub for tests and no-key runs.

Continued on next page

Team member	Primary contributions
RATHNAYAKE R.M.D.D. (220525K)	Led data preparation, evaluation, validation and external co-ordination. Handled communication with The Kingsbury for case-study coordination and project context, helping the team align the prototype with a real hotel operations setting. Prepared connector-shaped fixture datasets and real-review transforms, validated source identity and import consistency, and helped generate specificity-preserving incident data so the pipeline could be tested on concrete complaints rather than vague categories. Verified the reported corpus and outcome figures, including 2,939 analysed reviews, 2,294 negative/mixed reviews, 117 active issues, 587 emerging signals, 77% complaint linkage and the top issue counts. Documented data-source policy, review authenticity caveats, evaluation limits, source-risk handling and strategic risk mitigations used in the report and presentation.
WIJETHUNGA W.M.D.P. (220727G)	Led the frontend implementation and presentation material. Built and refined the Next.js/React operator interface as a calm, credible internal hotel operations tool rather than a marketing page, following the product principles of keeping evidence close to decisions and making ownership explicit. Worked on the dashboard KPIs, sentiment/risk visualisations, review browsing, issue queues, department and priority filters, active/emerging issue views, evidence badges, assignee/status visibility and demo usability. Prepared the visual materials for the final delivery, including screenshots, presentation flow, speaker notes and slide content explaining the problem, pipeline, findings, ROI, competitive edge and risk mitigations.

In addition to these primary areas, final integration, testing, report review and presentation rehearsal were shared responsibilities. The team jointly checked that the delivered system could be run locally with PostgreSQL, FastAPI and Next.js; that local NLP models, Gemini-based issue detection and the deterministic stub mode matched the intended product boundary; that review ingestion and detection produced the reported 2,939-review / 117-active-issue outcome; and that the final report, presentation and risk analysis remained consistent with the source-code implementation and project documentation.

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